INCH-POUND
MIL-M-38510/656
AMENDMENT 3
23 May 2000
SUPERSEDING
MIL-M-38510/656
AMENDMENT 2
31 March 1992

MILITARY SPECIFICATION
MICROCIRCUITS, DIGITAL, HIGH SPEED, CMOS FLIP-FLOPS,
MONOLITHIC SILICON

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/656, dated 31 March 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2.3 Case outlines: Delete in its entirety and substitute with the following:

Outline letter	Package descriptive designator (see MIL-STD-1835)
R	GDIP1-T20 or CDIP2-T20, 20-lead dual-in-line package
S	GDIP2-F20 or CDIP3-F20, 20-lead flat package
X	See figure 5
2	CQCC1-N20, 20 terminal leadless chip carrier package

*1.3 Thermal resistance, delete in its entirety and substitute with the following:
"Thermal resistance, junction-to-case (θ_{JC}) - - - - - - - - - See MIL-STD-1835".

PAGE 3

*3.2.4 Delete and substitute as; Case outlines. The case outlines shall be as specified in 1.2.3 and as specified on figure 5.

PAGE 7

Table I: I_{OZL} and I_{OZH}, add device type 52 to the device type column. In the max limits column of I_{OZL} delete "-0.2 V" and substitute "2.0". In the max limits column of I_{OZH} delete "0.2 V" and substitute "2.0".

PAGE 13

Figure 1: Delete "Case R" and substitute "Cases R and S". Delete "DUAL-IN-LINE-PACKAGE".

PAGES 15 THROUGH 23

Figure 1: Delete "DUAL-IN-LINE-PACKAGE" and substitute "Cases R and S".

PAGE 19

*Figure 1: Delete "Cases R and S" and substitute "Cases R, S, and X".

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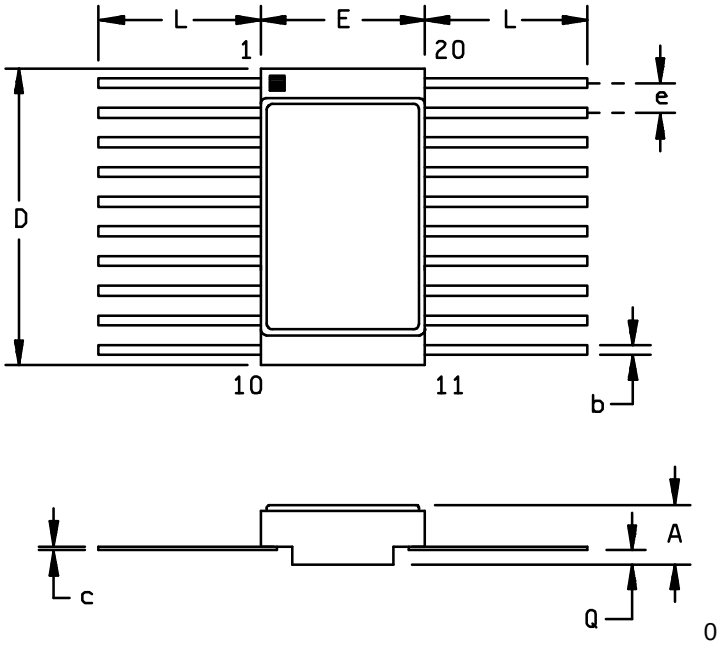
*Add case outline X as FIGURE 5.

"Device type 04

Case outline X



Device type 04, case outline X						
Symbol	Inches			Millimeters		
	Min	Nom	Max	Min	Nom	Max
A	.045		.085	1.14		2.16
b	.015		.019	0.38		0.48
c	.003		.006	0.076		0.152
D	.505		.515	12.83		13.08
E	.275		.285	6.99		7.24
e	0.045		0.055	1.14		1.40
L	.250		.370	6.35		9.39
Q	.010			0.25		
N	20			20		

FIGURE 5. Case outline X."

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PAGES 32 THROUGH 53

Table III: Delete "Cases R, 2" and substitute "Cases R, S, 2".

PAGE 36

Table III, device type 02, I_{OZH}, T_C = +25°C in the max limits column, delete "0.1" and substitute "0.2".
Table III, device type 02, I_{OZL}, T_C = +25°C in the max limits column, delete "-0.1" and substitute "-0.2".
Table III, device type 02, I_{OZH}, T_C = +125°C in the max limits column, delete "0.2" and substitute "2.0".
Table III, device type 02, I_{OZL}, T_C = +125°C in the max limits column, delete "-0.2" and substitute "-2.0".

PAGES 42 THROUGH 45

*Table III: Delete "Cases R, S, 2" and substitute "Cases R, S, X, and 2".

PAGE 43

Table III, device type 04, I_{OZH}, T_C = +25°C in the max limits column, delete "0.1" and substitute "0.2".
Table III, device type 04, I_{OZL}, T_C = +25°C in the max limits column, delete "-0.1" and substitute "-0.2".
Table III, device type 04, I_{OZH}, T_C = +125°C in the max limits column, delete "0.2" and substitute "2.0".
Table III, device type 04, I_{OZL}, T_C = +125°C in the max limits column, delete "-0.2" and substitute "-2.0".

PAGE 47

Table III, device type 05, I_{OZH}, T_C = +25°C in the max limits column, delete "0.1" and substitute "0.2".
Table III, device type 05, I_{OZL}, T_C = +25°C in the max limits column, delete "0.1" and substitute "-0.2".
Table III, device type 05, I_{OZH}, T_C = +125°C in the max limits column, delete "0.2" and substitute "2.0".
Table III, device type 05, I_{OZL}, T_C = +125°C in the max limits column, delete "0.2" and substitute "-2.0".

PAGE 51

Table III, device type 06, I_{OZH}, T_C = +25°C in the max limits column, delete "0.1" and substitute "0.2".
Table III, device type 06, I_{OZL}, T_C = +25°C in the max limits column, delete "-0.1" and substitute "-0.2".
Table III, device type 06, I_{OZH}, T_C = +125°C in the max limits column, delete "0.2" and substitute "2.0".
Table III, device type 06, I_{OZL}, T_C = +125°C in the max limits column, delete "-0.2" and substitute "-2.0".

PAGES 54 THROUGH 57

Table III: Delete "Case R" and substitute "Cases R, S, 2".

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PAGE 55

Table III, device type 52, I_{OZH}, T_C = +25°C in the max limits column, delete "0.1" and substitute "0.2".
Table III, device type 52, I_{OZL}, T_C = +25°C in the max limits column, delete "-0.1" and substitute "-0.2".
Table III, device type 52, I_{OZH}, T_C = +125°C in the max limits column, delete "0.2" and substitute "2.0".
Table III, device type 52, I_{OZL}, T_C = +125°C in the max limits column, delete "-0.2" and substitute "-2.0".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA-CC

Preparing activity:
DLA-CC

Review activities:
Army - AR, MI, SM
Air Force - 19, 99
NAVY - AS, CG, MC, SH

(Project 5962-1868)

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